

Supplementary Material

GUIDE: Reinforcement Learning for Behavioral Action Support in Type 1 Diabetes

Saman Khamesian et al.

1 Experimental Configuration

The following tables report the hyperparameters and settings used in the GUIDE experiments.

Table S1. Action-space bounds used in the GUIDE experiments.

Action Component	Range
Carbohydrate intake ($[c_{\min}, c_{\max}]$)	5–50 g
Proposed bolus insulin ($[i_{\min}, i_{\max}]$)	2–15 U

Table S2. Algorithm-specific hyperparameters used in GUIDE; SAC-Offline and SAC-Online use the same configuration.

Factor	Setting
<i>TD3-BC</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
Behavior cloning coefficient (α_{BC})	1.5
Policy noise (σ)	0.2
Noise clip (c)	0.5
Actor learning rate (η_{π})	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>CQL-BC</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
CQL regularization coefficient (α_{CQL})	0.05
Behavior cloning coefficient (α_{BC})	2.5
Actor learning rate (η_{π})	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>SAC-Offline and SAC-Online</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
Entropy coefficient (α)	0.2
Actor learning rate (η_{π})	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>PPO</i>	
Discount factor (γ)	0.99
Learning rate (η)	3×10^{-4}
GAE parameter (λ)	0.95
Clip ratio (ϵ)	0.2
Entropy coefficient (β)	0.001
Value loss coefficient (c_v)	0.5

Table S3. Bolus-safeguard parameters used during GUIDE training and evaluation.

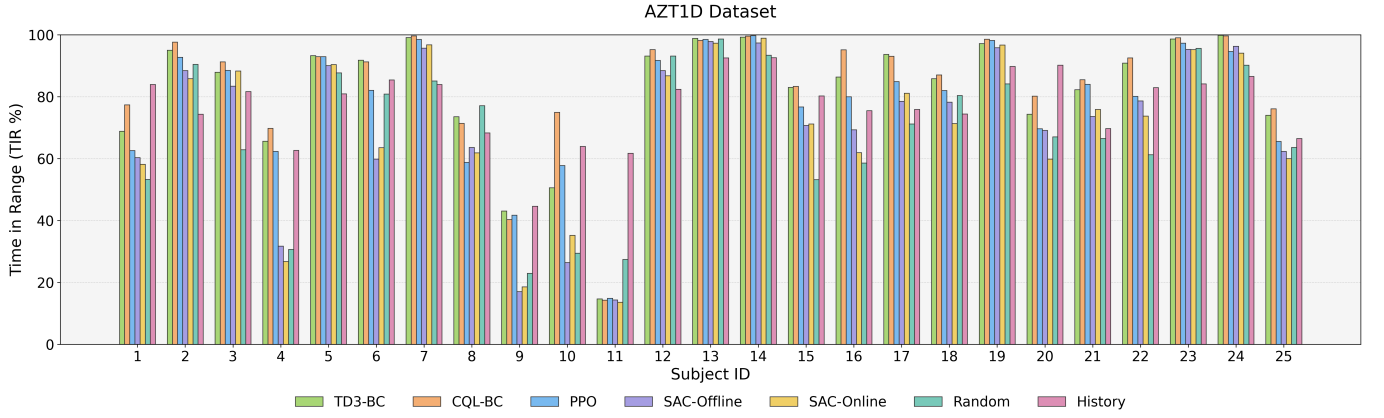
Parameter	Setting
Active-insulin duration	4 h
Maximum active insulin ($I_{\max}$)	30 U
Cumulative-bolus limit ($B_{\max}$)	60 U
Cumulative-bolus window (W)	4 h
Minimum bolus interval	2 h
Proposed bolus range	2–15 U
Minimum executable bolus	2 U

Table S4. Action-dependent reward functions and parameters used in the GUIDE experiments. Here, $\bar{g}_t$ is the mean of the 12 predicted CGM values over the upcoming hour; Δ_t^m and Δ_t^i denote the elapsed time since the previous meal and insulin administration, respectively, in five-minute intervals; q_t denotes the predicted CGM slope; c_t is the number of consecutive occurrences of the same non-NOTHING action; and $[P]$ equals one when condition P is satisfied and zero otherwise.

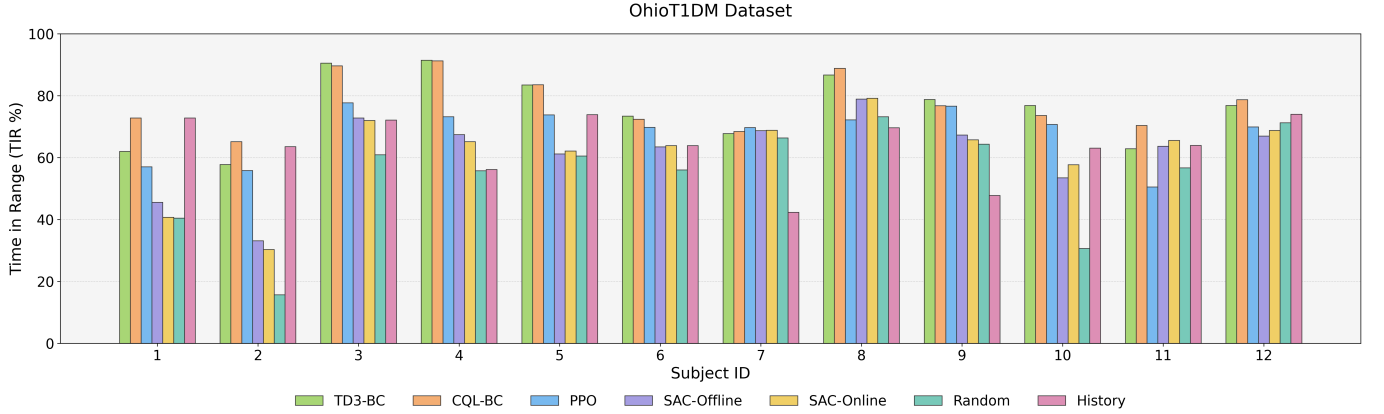
Function $f_j(s_t, a_t)$	α_j	$[L_j, U_j]$
<i>$a_t = \text{EAT}$</i>		
$\max(90 - \bar{g}_t, 0)$	5	$[0, 200]$
$-\lceil \bar{g}_t > 150, \Delta_t^m < 24 \rceil$	100	$[-100, 0]$
$\begin{cases} 1, & \Delta_t^i < 12, \\ 0.2, & \Delta_t^i \geq 12 \end{cases}$	25	$[5, 25]$
<i>$a_t = \text{INJECT}$</i>		
$-\max(150 - \bar{g}_t, 0)$	10	$[-500, 0]$
$\max(\bar{g}_t - 185, 0)$	20	$[0, 2000]$
$-\lceil \Delta_t^i < 24 \rceil$	100	$[-100, 0]$
$\lceil \Delta_t^m \leq 18, \bar{g}_t > 140, q_t > 0 \rceil$	75	$[0, 75]$
<i>Sleep</i>		
$2[a_t = \text{NOTHING}] - 1$	100	$[-100, 100]$
<i>Stability</i>		
$-\lceil a_t \neq \text{EAT} \rceil \max(90 - \bar{g}_t, 0)$	5	$[-200, 0]$
$-\lceil a_t \neq \text{INJECT} \rceil \max(\bar{g}_t - 185, 0)$	20	$[-2000, 0]$
$\lceil a_t = \text{NOTHING}, 120 < \bar{g}_t < 140 \rceil$	50	$[0, 50]$
$-\lceil a_t \in \{\text{EAT}, \text{INJECT}\} \rceil \max(c_t - 2, 0)$	950	$(-\infty, 0]$

2 Subject-Level TIR Comparison

Figure 1 presents the subject-level TIR results for the AZT1D and OhioT1DM datasets. For AZT1D, the historical mean TIR of 77.41% indicates generally favorable glycemic control across the cohort, although outcomes vary among subjects. Some subjects also have historical



(a) Subject-level TIR results for the 25 subjects in the AZT1D dataset.



(b) Subject-level TIR results for the 12 subjects in the OhioT1DM dataset.

Figure 1: Subject-level comparison of time in range (TIR) across the evaluated methods for (a) the AZT1D dataset and (b) the OhioT1DM dataset. For each subject, the five reinforcement learning algorithms—TD3-BC, CQL-BC, PPO, SAC-Offline, and SAC-Online—are compared with the Random policy and the subject’s historical TIR. The RL and Random results are averaged across the evaluation runs and random seeds.

TIR values above 90%, showing that the values near the upper limit achieved by some algorithms are consistent with the observed glycemic profiles of this cohort. For OhioT1DM, the historical mean TIR is lower at 63.62%, and the TIR achieved by the evaluated algorithms varies across subjects. The historical TIR of each subject is included as a contextual reference, and the plots provide the subject-level distributions underlying the cohort-level mean and standard deviation reported in the main results.

3 Contextual Comparison

The main manuscript compares the evaluated RL algorithms under the same personalized simulator, state representation, action space, reward formulation, and evaluation protocol. This controlled comparison showed that GUIDE with CQL-BC achieved the strongest overall glycemic performance among the algorithms evaluated within the proposed environment, with a TIR of $85.49 \pm 19.86\%$, TAR of $13.97 \pm 19.96\%$, and TBR of $0.53 \pm 1.33\%$. For broader context, Table S5 summarizes GUIDE alongside

representative closed-loop control and RL-based glucose-management approaches reported in the literature.

Most of the prior approaches summarized in Table S5 were trained and evaluated on the UVA/Padova physiological simulator [8] using synthetic data generated from virtual subjects. GUIDE, however, was evaluated within a personalized data-driven simulation framework based on a glucose prediction model trained on real-world patient data. Beyond differences in evaluation settings, prior approaches primarily focus on automated insulin delivery, where the agent controls basal and/or bolus insulin. GUIDE instead incorporates a broader behavioral action representation that includes bolus insulin, carbohydrate intake, and explicit timing decisions. Given these differences, the reported glycemic metrics, including TIR, TAR, and TBR, are provided only for contextual reference and should not be interpreted as direct head-to-head comparisons. Within this context, GUIDE achieves strong glycemic outcomes while operating under a broader behavioral action space than conventional insulin-delivery controllers.

Table S5. Contextual summary¹ of GUIDE (CQL-BC) and representative closed-loop control and RL-based glucose-management approaches reported in the literature. TIR, TAR, and TBR are reported as presented in the respective studies. RF denotes the reward formulation, Meal Ann. indicates whether meal announcements were incorporated, and Cohort Size (#N) denotes the number of evaluated subjects.

Algorithm	TIR (%)	TAR (%)	TBR (%)	RF	Action Support	Meal Ann.	Cohort Size (#N)
PID [1]	68.2	35.9	1.7	N/A	Basal + Bolus	Yes	30
SAC [1]	64.1	38.8	3.7	Exponential	Basal + Bolus	Yes	30
BCQ [2]	65.8	33.2	1.0	Magni	Basal	Yes	30
CQL [2]	56.2	43.7	0.1	Magni	Basal	Yes	30
TD3-BC [2]	65.3	34.5	0.2	Magni	Basal	Yes	30
Single PPO [3]	62.42	34.93	2.65	Magni	Basal + Bolus	Yes	10
Dual PPO [3]	69.3	18.71	1.99	Parabolic	Basal + Bolus	Yes	10
MBRL [4]	65.63	N/A	N/A	Kovatchev	Bolus	No	9
G2P2C [5]	72.69	24.10	1.21	Custom	Basal + Bolus	No	30
Deep RL [6]	77.36	22.45	0.19	Magni	Basal + Bolus	No	30
Bioinspired RL [7]	89.56	9.49	0.95	Homeostatic	Basal + Bolus	No	20
GUIDE (CQL-BC)	85.49	13.97	0.53	Custom	Bolus + Carb + Timing	Yes	25

¹ The studies differ in their datasets, simulators, cohort definitions, evaluation protocols, and control objectives. Accordingly, the reported results are provided only for contextual reference and do not constitute direct head-to-head comparisons.

References

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